

DRAM Notes

Memory organization

0 Banks of banks, each bank contains rows and columns of possible addresses.

0 In a transaction, a variable number of columns are (usually 8) accessed and saved into buffers, usually up to 128 bits. These are then ~~transferred~~ transferred as a burst, whose characteristics are determined by a command signal from the control.

DRAM Signals and Protocols

0 DRAM devices operate on a 4-stage protocol, which can be pipelined.

1. Command transport and decode

0 Signals move through registers, decode commands, ~~instructions~~ bank, row, and column number.

2. Memory is either fetched or written into the DRAM arrays.

3. I/O is gated and latched to proper drivers.

4. Data is received or sent.

Command format:

0 t_{cmd} = time to send command from controller to device.

0 $t_{parameter 1}$ = time command spends in a bank.

0 $t_{parameter 2}$ = time command spends in use of resources shared by all banks.

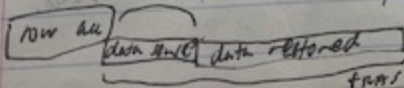
Row access commands:

0 Move data from a bank and then uses the sense amplifiers to restore data back into the cell.

0 Two timing parameters are applied here t_{rco} , t_{ras} .

0 t_{rco} (row column delay) time to move from DRAM cell to sense amplifiers

0 t_{ras} (row access strobe), data restored to arrays.



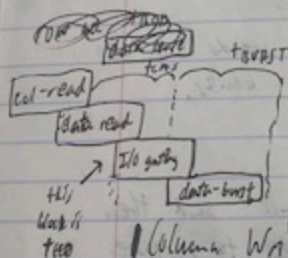
Column Read Command: moves data from sense amplifiers. Happens during t_{RAS} interval while sense amplifiers hold their values.

t_{RCAS} : time from column read command to data placed on bus.

t_{CCD} : internal burst length; column to column time between bursts, caused by I/O gating.

t_{BURST} : timing of burst, caused by protocols.

t_{BURST} has to be greater than t_{CCD} so data is properly outputted per beat.



Column Write: moves data from memory controller to sense amplifiers. During t_{RAS}

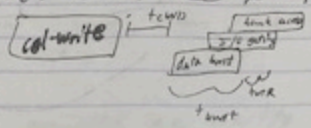
t_{CWD} : time from ~~command~~ command bus to data bus for ~~data~~ data.

t_{CWE} : time for write data to propagate into DRAM arrays.

t_{CWR} : time for I/O gating to switch from write to read.

t_{WR} must be respected before precharge commands.

t_{WTR} must be respected before ~~write~~ read commands.



Precharge Commands

Resets sense amplifiers and bitlines for a given DRAM array.

t_{RP} is time from command assertion to precharging bitlines and sense amplifiers. Must be respected before subsequent RAS commands.

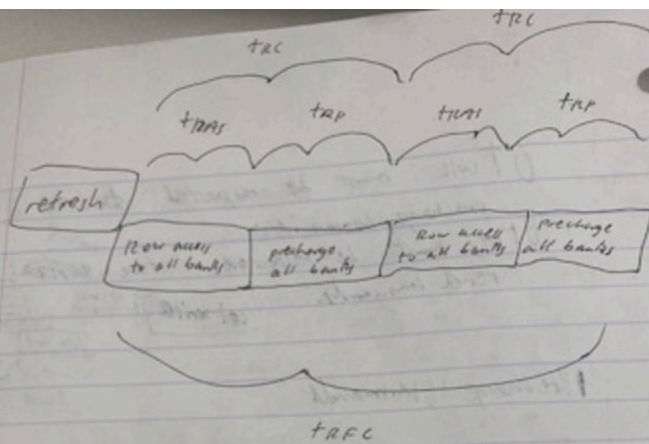
$$t_{RC} = t_{RAS} + t_{RP}$$

Time it takes to bring data to sense amplifiers, restore data to cells and precharge bitlines to reference voltage for another row access command.

Refresh

Refreshes one row in each bank.

When command issued row number is registered in DRAM device, and then sent to all banks to refresh. One row refresh takes t_{RFC}



Read cycle:

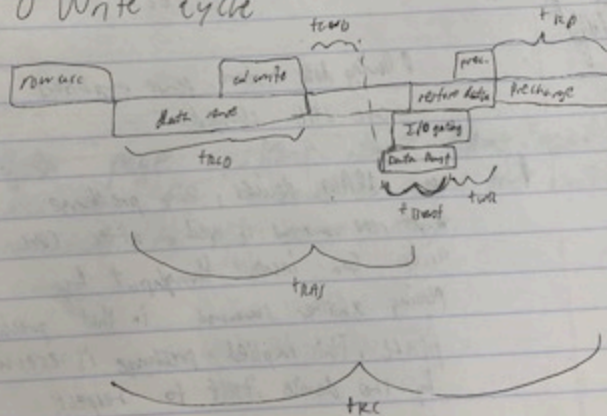
- 0 Row access brings thousands of bits from a row, the column access brings hundreds of bits from adjacent columns.
- 0 Caching thousands of bits using sense amplifiers for each row helps incur less latency from additional row accesses by applying principle of locality.
- 0 Data is brought from DRAM cells to sense amplifiers after t_{RP}

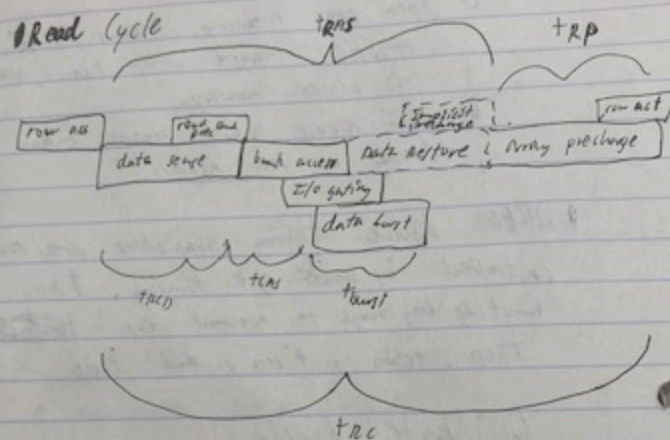
○ Open page memory, keeping sense amplifiers charged after t_{RAS} , opposite is closed memory.

○ Column access can occur many times during a row access.

● DRAM device timing parameters are mostly controlled by write gate timing. t_{RAS} must be long enough to account for ~~t_{RAS}~~
 t_{RCD} , t_{CWB} , t_{CCD} , and t_{WR}

○ Write cycle





During data restore, sense amplifiers are still charged.

In some DRAM devices, a precharge and read command is used after row access can increase throughput by placing another command in that precharge's place. This implied precharge is executed by the device itself to respect t_{RAS} .

Don

More modern DRAM devices have
a posted CAS command, which is
an input CAS after an initial
CAS. Encoded in the command
is a delay time to respect t_{RCD} .

These input commands
simplify DRAM controller.

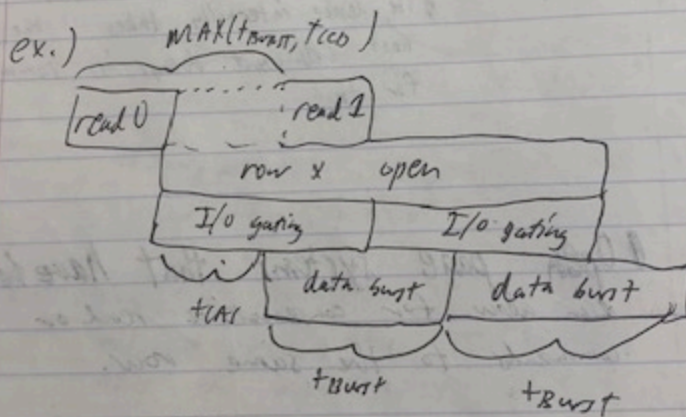
t_{RCD} is the delay requirement.

The device internally takes the
next CAS, but delays its command
for t_{RCD} .

Open page systems that have locality
can allow for consecutive read or write
commands to the same row.

Scheduling Read and Write transactions

- Commands, especially those to the same row of same banks, can be pipelined. Column read commands of the same row on different banks can be pipelined consecutively.



- Pipelined read instructions.

1 $t_{RP} + t_{CB} = \text{time to precharge}$
or $t_{burst} - t_{AP} + t_{RP}$ after column access command.

2 Consecutive ~~reads~~ reads to different rows of same bank.

0 time from read 0 to read 1:

$$t_{burst} + t_{RP} - t_{CB} + t_{RP} + t_{CB}$$

total length
from RAS to
precharge

precharge
and another
RAS.

0 This also assumes that
 t_{RAS} passes before next row
access. If not, controller must
wait until t_{RAS} fully passes.

②

1 Without command reordering,
consecutive reads to different
rows is a minimum of $t_{RP} + t_{ACD}$

With reordering, a precharge command
can be issued to a different
bank during a column read
command.

min timing between commands
is now $t_{RP} + t_{PCP} - t_{CMD}$
for transactions to different
banks.

Other CMD issued in
parallel is another
bank precharge. This happens
during first command, not after.

t_{RRAS} is for rank
utilization when accessing
different ranks

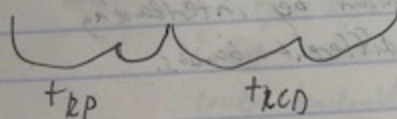
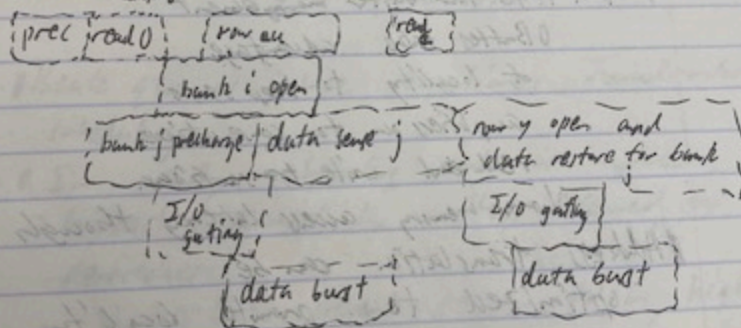
• Bank conflict to same bank for write:

$$- t_{\text{wp}} + t_{\text{burst}} + t_{\text{wr}} + t_{\text{rp}} + t_{\text{rd}}$$

Assumes $t_{\text{wp}} + t_{\text{burst}} + t_{\text{wr}}$ is greater than t_{ras} .

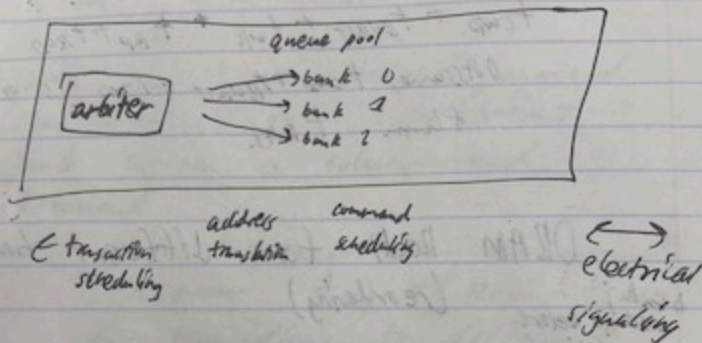
DRAM Reads for different banks (reordering)

bank j precharge before reading from bank i



$$t_{\text{rp}} + t_{\text{rd}} = t_{\text{amb}}$$

DRAM Controller Overview



1 open-page row-buffer management

0 Buffer take advantage of locality to keep some amplifiers on for consecutive read and write transactions.

0 Low memory access latency through

Address translation can be optimized to promote locality, or parallelism by interleaving access to different banks.

0 The highest bits in address are typically mapped to the rows within bank to increase row hit chance.

0 Column addresses are mapped to lower bits of the address for spatial locality.

0 Banks are mapped to lower bits as well to allow for a degree of interleaving and parallelism.

• Bank queues use sometimes round-robin scheduling.

• In priority schedules, reads are given highest, followed by writes, followed by refreshes.

0 To prevent starvation, high priority agents can instead be called low-latency, while lower bandwidth or guaranteed bandwidth can be scheduled fairly and inserted as necessary.